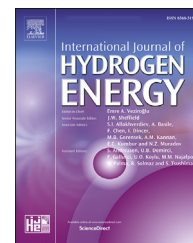


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A new cost-minimizing power-allocating strategy for the hybrid electric bus with fuel cell/battery health-aware control

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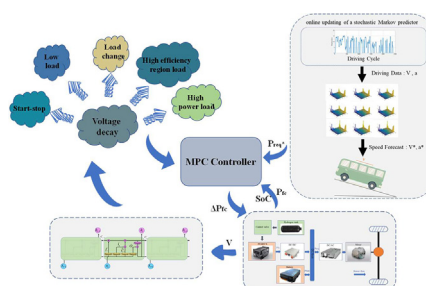
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HIGHLIGHTS

- A PEMFC aging model considering the full efficiency region is established.
- An EMS considering fuel cell/battery health awareness is developed for hybrid bus.
- The LSTM is used to accurately predict the remaining life of on-board fuel cell.
- An improved online Markov method is developed to predict vehicle working conditions.
- The effectiveness of EMS with fuel cell/battery health awareness is verified.

GRAPHICAL ABSTRACT



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ABSTRACT

The two primary challenges preventing the commercialization of fuel cell hybrid electric vehicles (FCHEV) are their high cost and limited lifespan. Improper use usage can also hasten the breakdown of proton exchange membrane fuel cell (PEMFC). This paper proposes a new cost-minimizing power-allocating technique with fuel cell/battery health-aware control to optimize the economic potential of fuel cell/battery hybrid buses. The proposed framework quantifies the fuel cell (FC) deterioration of the whole working zone in a real hybrid electric bus using a long short-term memory network (LSTM), which reduces the time required to get the key lifetime parameters. A new FC lifespan model is embedded into the control framework, together with a battery aging model, to balance hydrogen consumption and energy source durability. In addition, in the speed prediction step, an enhanced online Markov prediction approach with stochastic disturbances is presented to increase the forecast accuracy for future disturbances. Finally, comparative analysis is used to verify the efficacy of the suggested approach, which shows that when compared to

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the benchmark method, the proposed strategy may extend the FC lifetime and lower operating costs by 5.04% and 3.76%, respectively.

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Introduction

With the shortage of fossil fuels and increasing environmental pollution, renewable energy generation is booming around the world [1]. PEMFC can convert the chemical energy in hydrogen into electricity. Compared with other fossil fuels generating power, PEMFC has the advantages of compact size, high efficiency and low operating temperature, which makes fuel cell vehicles gradually come into public vision [2–4]. With the advantages of zero emissions, short charging time and long driving distance, fuel cell hybrid electric vehicles (FCHEVs) are considered to be one of the most promising sustainable energy vehicles for future transportation systems [5]. However, compared with conventional vehicles, the fuel cell-battery-motor coupling system unique to fuel cell hybrid electric vehicles is quite complex [6], and the performance of the fuel cell and battery system is highly susceptible to operating conditions during use, and improper loading not only shortens its lifetime [7], but also increases vehicle operating costs due to hydrogen consumption [8]. To address the above issues, this paper designs a novel energy management strategy (EMS) with fuel cell/battery health awareness to maximize the economic potential of fuel cell hybrid vehicles.

Currently, many different EMSs have been developed for FCHEVs, and they can be broadly classified into rule-based and optimization-based strategies [9]. Rule-based policies center on heuristic rules governing vehicle operating modes to distribute power demand based on a set of deterministic or fuzzy rules. For example, when the battery state-of-charge (SOC) is above the upper threshold, the pure electric mode is used; when the battery SOC is below the lower threshold, the fuel cell system operates at the most efficient point or the rated point [10]. Although such methods have low computational complexity, they rely excessively on engineering experience and are still a long way from the optimal vehicle control objectives. Optimization-based strategies could significantly improve the performance of FCHEV compared to rule-based strategies [11]. This type of method is characterized by converting the energy management objectives into a mathematical optimization problem given a hybrid powertrain model, and obtaining the control sequence under multiple equality or inequality constraints. Dynamic programming (DP), as a globally optimal search algorithm in the decision-making process, is often used to develop theoretically optimal energy management strategies [12]. Xu et al. used DP to allocate power between the two power sources to minimize the operating costs of fuel cell hybrid electric vehicle [13]. However, the computational cost of DP is very high, especially when the dynamic models have multiple states or control variables [14]. In addition to DP, other global optimization algorithms, genetic algorithms (GA) [15] and

convex programming (CP) [16], can also be used to solve the optimal power allocation problem for FCHEV. However, their control effects can only be achieved offline due to the need for a priori knowledge of future driving conditions [17].

To explore the possibility of online optimization of EMS, a variety of control strategies have been developed, e.g., Equivalent Consumption Minimization Strategies (ECMS) [18,19], Pontryagin's Minimum Principle (PMP) [20–23] and Model Predictive Control (MPC) [24–26]. Wang et al. [27] estimated the boundary power of the maximum efficiency range (MER) by the forgetting factor recursive least squares online identification method, and then solved the optimization problem of the equivalent consumption minimization strategy by sequential quadratic programming, and the experimental results showed that the proposed optimized ECMS optimized the system hydrogen consumption and helped to improve the fuel cell performance compared with the rule-based EMS. Zheng et al. [28] used the PMP algorithm to extend the fuel cell life by limiting the power change rate of the fuel cell. Hu et al. [29] constructed a battery health-aware EMS for FCHEVs equipped with hybrid energy storage by adding a battery state-of-health (SOH) model to an efficient CP framework. Zhou et al. [30] developed an FCHEV energy management strategy for perceiving fuel cell and battery system lifetime based on the MPC framework to minimize vehicle operating costs.

Although the above studies proved the potential ability of online optimization of EMSs, most of them only take the cost of hydrogen consumption as the main optimization objective and do not focus on the effect of fuel cell systems (FCS) operating conditions on fuel cell degradation. This leaves the economic potential of FCHEVs under-evaluated. A few papers attempted to establish EMSs cognizant of the tradeoff between hydrogen economy and fuel cell degradation. For example, in Ref. [25], except for the saving in hydrogen consumption, only the degradation of the FCS caused by load change was studied. However, other FCS operating conditions that can affect the FCS's lifetime were not considered. Even though few studies [30,31] have incorporated the cost of fuel cell/battery degradation into the MPC cost function, the degradation effects of FCS mainly considered the specific conditions for example, the start-stop, load change, high load, etc. As these specific conditions have relatively high impact on the FC health. However, the normal working condition mainly operated in the FC high efficiency region is ignored. Considering that in the actual operation of FCHEVs, the fuel cell operates in the high efficiency region for a very high percentage of time, and it is actually the essential characteristics of a good energy management strategy. Therefore, it is necessary to develop new FC lifetime model that with fully consideration of FC working regions to achieve fuel cell

health-aware control in the hybrid electric vehicle application. Fuel cell degradation is generally modeled on the stack level. The main fuel cell modeling methods that have been applied in vehicles are Stack voltage degradation model, Impedance estimation based on Electrochemical impedance spectrometry and Remaining useful life estimation [32]. Considering the complexity of the degradation process of proton exchange membrane fuel cells (PEMFC) in vehicle applications, we choose stack voltage degradation model. However, it is time consuming and costly to obtain stack degradation data in real road tests. Therefore, a data-driven approach based on the long short-term memory network (LSTM) is proposed in this paper to obtain fuel cell lifetime model parameters more quickly and accurately.

It should be noted that the performance of MPC, compared to ECMS and PMP, is closely related to the prediction accuracy of future perturbations (e.g., vehicle speed). Therefore, in order to obtain better MPC control performance, a reliable speed forecasting method needs to be designed. Although many speed forecasting methods have been successfully developed, such as Markov chain (MC) based stochastic models [33,34] and neural network (NN) based machine learning methods [35,36], there is still much room to improve the prediction accuracy. Due to its simplified mathematical description and easy integration with optimal control problems, MC based velocity prediction methods have been successfully applied to stochastic control problems.

This paper aims to bridge these research gaps, and proposes a new cost-minimizing power-allocating strategy for the fuel cell hybrid electric bus (FCHEB). The main contributions of the paper include the following:

- FC degradation of the whole efficiency region in a real hybrid electric bus is quantitatively analyzed based on a data-driven approach and the remaining life of an on-board fuel cell is accurately predicted
- The whole working regions of both the fuel cell and battery are considered in the proposed real-time energy management method, and the aging models for both the battery and FC are embedded into the decision framework to balance hydrogen consumption and energy source durability.
- An improved online learning augmented (stochastic) Markov method is also developed to predict the working conditions for the hybrid electric bus as the key input for the proposed EMS.

The rest of this paper is as follows. Section 2 describes the fuel cell/battery hybrid system of FCHEB in detail. Section 3 describes the energy management strategy with fuel cell/cell health-aware control and the improved online update algorithm for the transition probability matrix (TPM). Section 4 describes other benchmarking strategies and discusses the results. Section 5 summarizes the conclusions.

System description

The configuration of the Yutong FCHEB is shown in Fig. 1. The PEMFC system is connected to the DC bus by a unidirectional DC-DC converter, with the purpose of regulating fuel cell

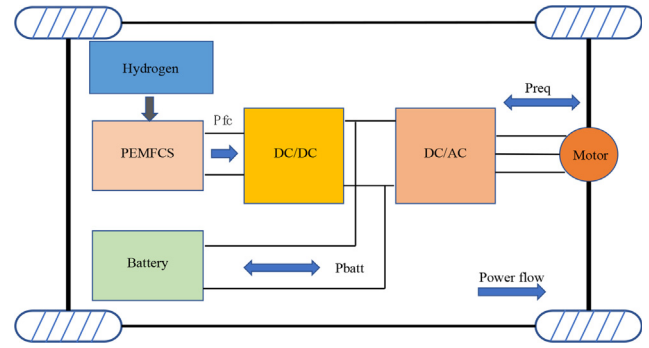


Fig. 1 – Configuration of the fuel cell hybrid bus.

current towards the DC bus, while the battery is straight connected to the DC bus. Since the vehicle is not equipped with the plug-in property, it can only be recharged via DC bus by PEMFC or by the regenerative energy [30]. The main parameters of each component are listed in Table 1.

Hybrid powertrain model

When the vehicle is driving on a non-horizontal road (Fig. 2) the traction force F_t can be calculated by formula (1), and the required power on the DC bus P_{req} is provided by fuel cell output power P_{fc} and power battery output power P_{bat} , and the power balance relationship can be expressed by (2) (3).

$$F_t = c_r mg \cos(\alpha) + mg \sin(\alpha) + \delta ma + \frac{1}{2} \rho A c_D v^2 \quad (1)$$

$$P_{req} = \frac{F_t \cdot v}{\eta_{bus}} = P_{fc} \cdot \eta_{DC/DC} + P_{bat} \quad (2)$$

$$\eta_{bus} = \eta_{DC/AC} \cdot \eta_{EM} \cdot \eta_{tra} \quad (3)$$

where m represents the vehicle's mass; v is the vehicle's velocity; c_r is the coefficient of rolling resistance; ρ is the ambient air density; A is the vehicle's front surface area; c_D is the aerodynamic coefficient; Moreover, η_{bus} , $\eta_{DC/DC}$, $\eta_{DC/AC}$, η_{EM} and η_{tra} denote the efficiency of FCHEB, DC-DC converter, DC-AC inverter, electric machine and driveline, respectively. In this paper the gravitational acceleration g is taken as 9.81 m/s^2 and the road slope α is assumed as zero.

Fuel cell model

- 1) Hydrogen Consumption Model: We adopt a 310-cell PEMFC system with the rated power at 60 kW for an electric bus. Its hydrogen mass consumption m_{H_2} can be calculated by Ref. [37]:

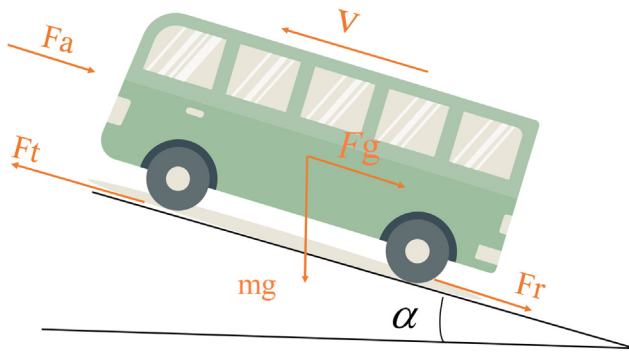
$$m_{H_2} = \int_0^t \frac{P_{fc}}{\eta_{fc} \cdot \rho_{H_2}} dt \quad (4)$$

$$\eta_{fc} = \frac{P_{fc}}{P_{H_2}} \quad (5)$$

where ρ_{H_2} is the chemical energy density of H_2 (MJ/kg); P_{fc} is the net power of the fuel cell system; η_{fc} is the efficiency of the PEMFC system; P_{H_2} is the theoretical power provided by

Table 1 – Main parameters of the FCHEB.

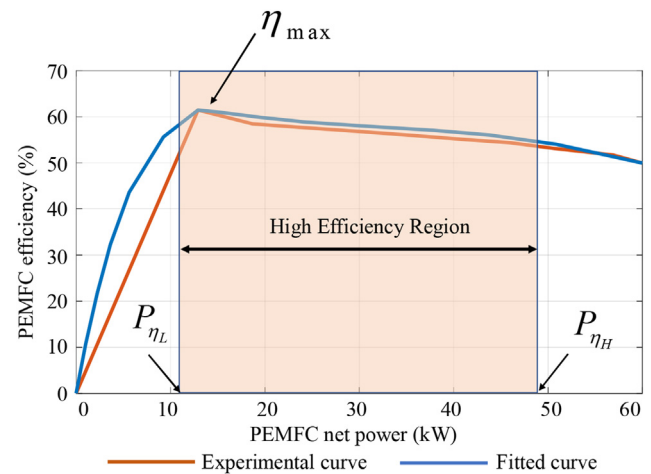
Component	Description	Value
Vehicle	Weight	13,500 kg
	Frontal surface area	8.16 m ²
	Rolling radius	0.466 m
	Rolling Resistance coefficient	0.0085
	Drag coefficient	0.55
PEMFC System	rated power	60 kW
Li-Ion Battery	Nominal Energy capacity	47.3 kW h
	Battery Capacity	90 A h
Electrical Machine	Max torque/speed	1800 Nm/5000 rpm
Transmission System	Main reduction ratio	6.2
DC/DC converter	Constant Efficiency	0.9
DC/AC inverter	Constant Efficiency	0.95

**Fig. 2 – Vehicle's dynamics in motion.**

H₂. The efficiency curve of the studied 60-kW PEMFC system is specified in Fig. 3. Note the peak efficiency $\eta_{\max}$ corresponds to the most efficient operating point of the PEMFC system. Besides, the orange area is defined as the high efficiency region of the studied PEMFC system. It will remain above 54% when $P_{fc} \in [P_{\eta_L}, P_{\eta_H}]$.

Many PEMFC lifetime tests have shown the performance degradation is linear with time. However, practical experiments over long periods of time will be costly and often obtain lifetime results too late to keep up with advances in fuel cell technology [41]. Therefore, realizing the PEMFC lifetime rapid prediction is necessary.

Considering that the PEMFC are complex multi-physics and multi-scale systems, the aging process of PEMFC is very complicated. We choose the stack voltage as the indicator of the degradation process [42]. In this work, we propose an innovative fuel cell health model based on the FCHEB using LSTM recurrent neural network. LSTM can realize the accurate prognostics of the remaining lifetime of PEMFC by learning the degradation data with non-linearities. This method does not need to deeply understand the degradation mechanisms of PEMFC, and thus is easy to implement with high computational efficiency. Therefore, it can be used in the online prognostic's applications. The LSTM neural network (NN) as shown in Fig. 4. Compared with traditional RNN architecture, an input gate, a forget gate and an output gate are added. So it can avoid gradient vanishing and gradient exploding problem [43–45]. x_t and x_{t-1} are the input of the current and previous time step, C_t and C_{t-1} are the cell state of the

**Fig. 3 – The efficiency curve of the PEMFC system.**

2) Health Model: The degradation of PEMFC system is a complex process involving multiple impact factors ranging from electrochemical to mechanical perspectives [38]. How to model the PEMFC system health evolution plays an important role in the economy-targeted power management of the bus. As reported in Ref. [39], the MEA degradation is mainly resulted from the three aspects: Catalyst layer degradation, Membrane layer degradation and Gas diffusion layer degradation. Numerous studies have shown that the reason why lifetime of automotive fuel cell is shorter than that of stationary fuel cell is the complex operation conditions undoubtedly, and so the effect on performance degradation when the fuel cell is operating in different working areas should be carefully considered [40]. In this study, when the PEMFC output power P_{fc} is less than or equal to 20% of the rated power is defined as low load, when the PEMFC output power P_{fc} is greater than or equal to 80% of the rated power is defined as high load, and the other PEMFC working areas are defined as high efficiency region load.

current and previous time step, h_t and h_{t-1} are the output of the current and previous time step, respectively.

First, the numerical value of the forget gate layer can be calculated by:

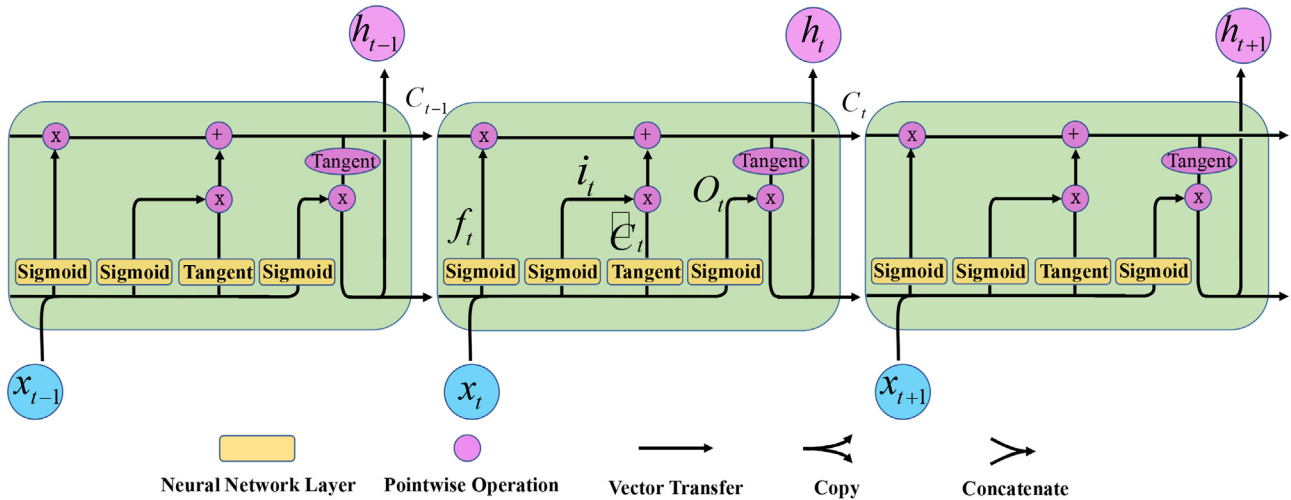


Fig. 4 – Architecture of the LSTM NN.

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \quad (6)$$

Then, the input gate layer and the candidate value $\tilde{C}_t$ can be obtained as:

$$\begin{aligned} i_t &= \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \\ \tilde{C}_t &= \tanh(W_c \cdot [h_{t-1}, x_t] + b_c) \end{aligned} \quad (7)$$

The current cell state can be expressed as:

$$C_t = f_t * C_{t-1} + i_t * \tilde{C}_t \quad (8)$$

Finally, the output gate layer can be obtained as:

$$\begin{aligned} O_t &= \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \\ h_t &= O_t * \tanh(C_t) \end{aligned} \quad (9)$$

where $\sigma(\cdot)$ is the activation function, $\tanh(\cdot)$ is the output function, W_f, W_i, W_c and W_o are the weight matrices, b_f, b_i, b_c and b_o are bias vectors.

The operating conditions of the PEMFC during the driving of the FCHEB mainly include start-stop, low load, load change, high efficiency region load and high load. The rated power of the PEMFC stack is 60 kW, and the rated current is 276 A.

Therefore, the fuel cell lifetime can be calculated by:

$$\begin{aligned} V_d &= N_{\text{cycle}}^1 v_1 + T_1 v_2 + N_{\text{cycle}}^2 v_3 + T_2 v_4 + T_3 v_5 \quad (a) \\ T_{\text{fuel}} &= \frac{\Delta V}{V_d} \quad (b) \end{aligned} \quad (10)$$

where N_{cycle}^1 represent the average start-stop numbers per hour; T_1 represent the average low load time per hour; N_{cycle}^2 is the average load change cycles per hour; T_2 is the average high efficiency region load operation time per hour; T_3 is the high load operation time per hour. V_d stands for the average rate of voltage degradation per hour while the vehicle driving. v_1, v_2, v_3, v_4, v_5 represents the voltage degradation rate under the corresponding operating conditions, respectively. $\Delta V = 0.07V$ is the allowable individual fuel cell voltage degradation value by the end of life (i.e. 10% voltage drop in fuel cell at rated current [46]) and T_{fuel} is the predicted PEMFC lifetime. The predicted result is shown in Fig. 5(a)-(b), the 90% voltage

data are used to train the LSTM neural network, and the rest of the data is used to verify the accuracy of the prediction.

As shown in Fig. 5(a) and (b), the trained LSTM shows very good prediction results, and the overall root mean square error (RMSE) is around 0.00088 V. For the prediction phase, the mean absolute error (MAE) is about 0.000687 V. The operating time of PEMFC under different operating conditions was counted and obtain the lifetime model of PEMFC by applying Eq. (10). The voltage degradation rate under different operating conditions is shown in Table 2.

Battery model

This paper adopts a 47.3 kW h lithium-ion battery pack as the energy storage unit in the hybrid powertrain. The equivalent circuit of battery model is illustrated in Fig. 6(a), where the DC bus voltage U_{DC} , the battery current I_{bat} and the state-of-charge SoC can be respectively calculated by (11)-(13).

$$U_{DC} = U_{ocv} - I_{bat} \cdot R_{bat} \quad (11)$$

$$I_{bat} = \frac{U_{ocv} - \sqrt{U_{ocv}^2 - 4R_{bat}P_{bat}}}{2R_{bat}} \quad (12)$$

$$\text{SoC}(t+1) = \text{SoC}(t) - \frac{I_{bat}(t)\Delta t}{Q_{bat}} \quad (13)$$

where $U_{ocv}, R_{bat}, P_{bat}, \Delta t, Q_{bat}$ are the open-circuit voltage, the internal resistance, the battery power, the sampling interval and the nominal battery capacity, respectively. During the charging and discharging process of the battery, its performance will gradually degrade. The fade of capacity is the important indicator of the battery aging. Because in practical applications, the operating conditions of the battery may be uncertain and unpredictable. The data-driven approach cannot describe the internal aging process. To estimate the capacity loss of the battery, a semi-empirical battery-aging model developed in Ref. [47] is adopted in this paper. The validity of the model has been verified in many previous studies and has been widely used for vehicular energy management. Its principle is briefly described as follows.

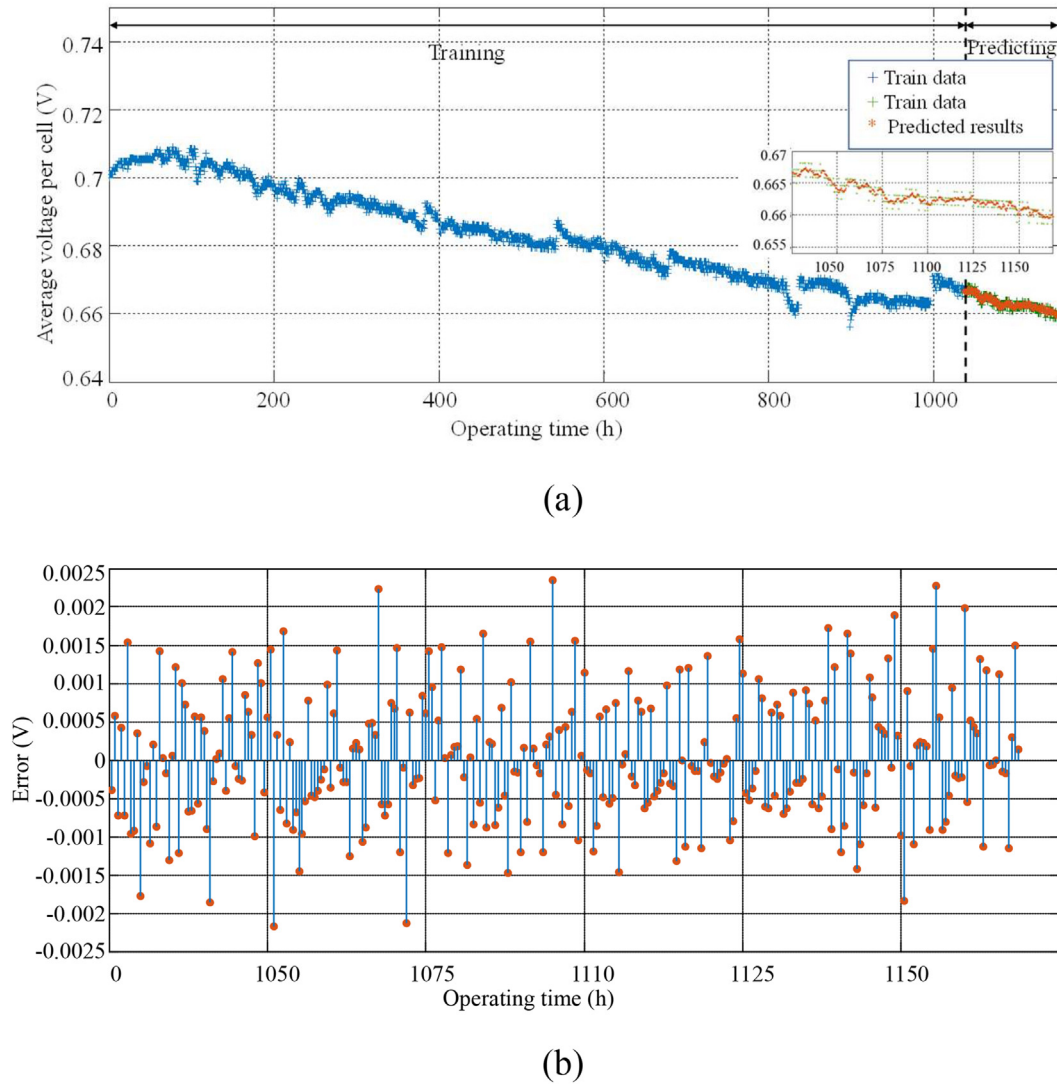


Fig. 5 – Voltage degradation curves for PEMFC.

Table 2 – Voltage Degrading rate under different operating conditions (cell level).

Operating conditions	Voltage degradation rate
Start-stop	$v_1 = 13.79 \mu\text{V}/\text{cycle}$
Low load	$v_2 = 9.42 \mu\text{V}/\text{h}$
Load change	$v_3 = 0.04234 \mu\text{V}/\text{kW}$
High efficiency region load	$v_4 = 4.881 \mu\text{V}/\text{h}$
High power load	$v_5 = 11.67 \mu\text{V}/\text{h}$

The battery capacity loss is defined as [47]:

$$\Delta Q = \alpha(c) \exp\left(\frac{-E_a(c)}{RT}\right) A(c)^z \quad (14)$$

where ΔQ is the percentage loss of battery pack capacity, c is the battery charge/discharge rate C-rate. α is the pre-exponential factor, which is a function of the C-rate, as showed in Table 3. Moreover, R is the ideal gas constant (8.31 J/mol), T is the battery cell temperature in K, A is the ampere-hour (Ah) throughput provided by battery during cycles, and

z is the power-law factor. The activation energy E_a are calculated by Ref. [47]:

$$E_a = 31700 - 370.3 \cdot c \text{ (J/mol} \cdot \text{K)} \quad (15)$$

z take 0.55. Note, when using (15) and Tables 3 and 1C ($c = 1$) corresponds to 2 A, where a de-rated 2 A h was used to simplify the experimentation. A 20% capacity fade is considered as the end-of-life (EOL) of an automotive battery pack [48], and thus the total Ah-throughput ($A_{\text{tol}}^{\text{EOL}}$) can be acquired by:

$$A_{\text{tol}}^{\text{EOL}}(c) = \left[\frac{20}{\alpha(c) \exp\left(\frac{-E_a(c)}{RT}\right)} \right]^{1/z} \quad (16)$$

Hence, the number of cycles N_{EOL} until the battery EOL can be attained by Ref. [49]:

$$N_{\text{EOL}}(c) = \frac{3600 A_{\text{tol}}^{\text{EOL}}(c)}{Q} \quad (17)$$

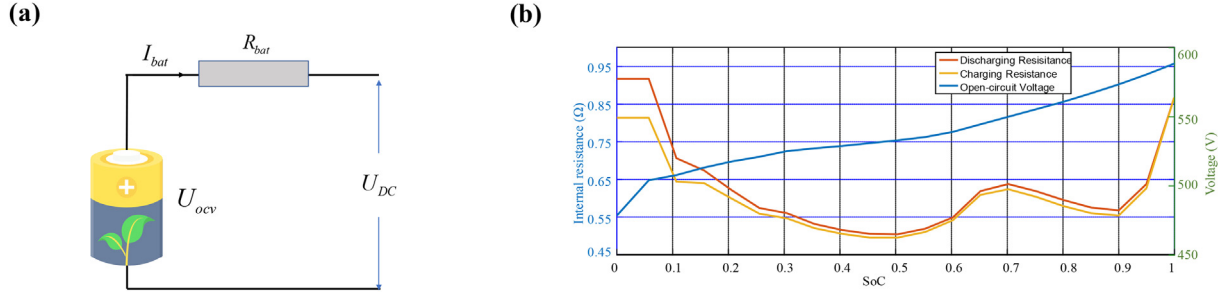


Fig. 6 – (a) Equivalent circuit of battery model; (b) Relationships among OCV, resistance and SOC.

Table 3 – PRE-EXPONENTIAL factor (α) as a function of battery current C-RATE [47].

C-rate	0.5	2	6	10
α	31,630	21,681	12,934	15,512

As a result, the state-of-health (SOH) is defined as follows [49]:

$$SOH(t) = 1 - \frac{\int_0^t |i(\tau)| d\tau}{2N_{EOL}(c)Q} \quad (18)$$

In discrete-time domain the corresponding dynamic SOH model can be expressed as [49]:

$$SOH(k+1) = SOH(k) - \frac{|i(k)|\Delta t}{2N_{EOL}(c)Q} \quad (19)$$

Then the dynamics of SOH under given C-rate, various temperatures can be obtained based on the SOH model.

Electric machine model

During vehicle running, the electric machine provides power and do regenerative braking. The efficiency map of the motor is plotted in Fig. 7.

$$\eta_{motor} = f(\omega_{motor}, T_{motor}) \quad (20)$$

where η_{motor} is the efficiency of the motor, ω_{motor} and T_{motor} are the speed and torque of the motor, respectively. Eq. (20) is usually given in the form of discrete 2-D look-up table. So, once the speed and torque requirements are determined, η_{motor} can be obtained from the look-up table extracted in Fig. 7.

EMS with fuel cell/battery health-aware control

This section introduces the design of the EMS with fuel cell/battery health-aware control and its schematic diagram shown in Fig. 8. At the supervisory layer, the Estimation of upcoming vehicle speed is achieved by the online updating of a stochastic Markov predictor. Then, the vehicle operating cost is quantified by MPC's objective function, and the hydrogen consumption cost and power cell and fuel cell degradation cost are considered. Finally, the optimal power allocation decision for each prediction interval is obtained by nonlinear solving.

Driving condition prediction

In the actual driving process, the demand power of the vehicle changes with the external environmental information and the driving habits of the driver, and its change law is unknown. Therefore, the change of demand power is a random process. In the future driving process, the demand power of the vehicle can be regarded as only related to the current speed and acceleration, but not to the historical information, so the process of changing the demand power of the vehicle can be considered as a Markov process.

Offline updating algorithm for TPM

A traditional finite-state Markov chain is a stochastic process that mainly satisfies two assumptions: (A1) The future state of the system is independent of the past and depends only on the current value. (A2) The possible values of the uncertain variables belong to a finite set, which is called the state space of the Markov chain [50]. We denote the state space of a finite state Markov chain as $S = \{x_j, j = 1, \dots, M\} \subset X \subset R^n$, where X is bounded and here we introduce the transfer probability.

$$p_{ij} = P(x^{k+1} = x_j | x^k = x_i) \approx \frac{N_{ij}}{N_{oi}} \quad (21)$$

where N_{ij} denotes the number of transfers from state x_i to state x_j in a given time interval, $N_{oi} = \sum_{j=1}^M N_{ij}$ is the total number

of transitions from state x_i in the same time. The state transfer matrix p can be formed from p_{ij} . The present and one step-ahead state probability vectors can be expressed as:

$$(\pi^{k+1})^T = (\pi^k)^T p \quad (22)$$

And for $t > 1$ steps ahead as [50]:

$$(\pi^{k+t})^T = (\pi^k)^T p^t \quad (23)$$

Online updating algorithm for TPM

According to (21), in a finite-state Markov chain, the transfer probability can be estimated based on the transfer frequency, then at moment k [50].

$$p_{ij} \approx \frac{N_{ij}(k)}{N_{oi}(k)} = \frac{N_{ij}(k)/k}{N_{oi}(k)/k} = \frac{F_{ij}(k)}{F_{oi}(k)} \quad (24)$$

where $F_{ij}(k)$ is the average frequency of $\sum_{t=1}^k f_{ij}(t)$ from state x_i to

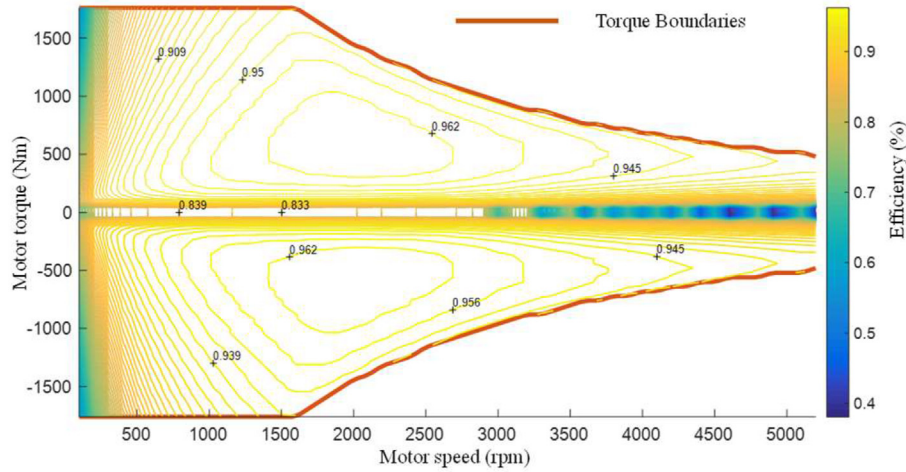


Fig. 7 – Map of motor efficiency.

state x_j and $F_{oi}(k)$ is the average transfer frequency of $\sum_{t=1}^k f_{ij}(t)$ from state x_i .

$$F_{ij}(k) = \frac{N_{ij}(k)}{k} = \frac{\sum_{t=1}^k f_{ij}(t)}{k} \quad (25)$$

$$F_{oi}(k) = \frac{N_{oi}(k)}{k} = \frac{\sum_{t=1}^k f_i(t)}{k} \quad (26)$$

If a transition from state x_i to x_j occurs at moment t , then $f_{ij}(t) = 1$; $f_i(t) = 1$ if a transition initiated from x_i at time instant t occurs; otherwise, these take zero values.

From equation (25) (26), a recursive formula for their online updating can be derived [50].

$$F_{ij}(k) = F_{ij}(k-1) + \frac{1}{k} [f_{ij}(k) - F_{ij}(k-1)] \quad (27)$$

$$F_{oi}(k) = F_{oi}(k-1) + \frac{1}{k} [f_i(k) - F_{oi}(k-1)] \quad (28)$$

By replacing $1/k$ with a constant $\delta \in (0, 1)$, (27), (28) will assign exponentially decreasing weights to the old data, allowing the recursive algorithm to be interpreted as an autoregressive model implementing an exponential smoothing algorithm with a forgetting factor δ . The recursive algorithm then becomes [51].

$$p_{ij}(k) = \frac{F_{ij}(k-1) + \delta [f_{ij}(k) - F_{ij}(k-1)]}{F_{oi}(k-1) + \delta [f_i(k) - F_{oi}(k-1)]} \quad (29)$$

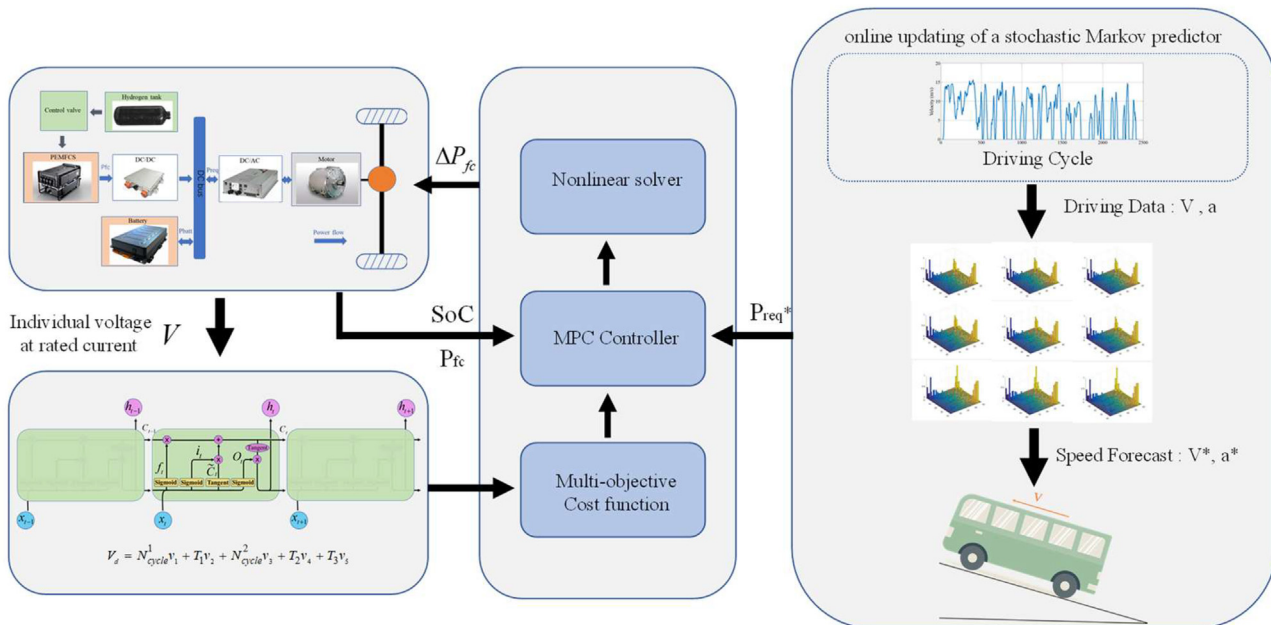


Fig. 8 – Proposed control framework of the EMS with fuel cell/battery health-aware.

The forgetting factor controls the update frequency $F_{ij}(k)$ and $F_{oi}(k)$ by assigning a set of exponentially decreasing weights to the old observations

$$W = [(1-\delta)^k \delta(1-\delta)^{k-1} \delta^2(1-\delta)^{k-2} \dots \delta] \quad (30)$$

Note that the sum of the elements of the vector W is equal to 1, which can be interpreted as a weighted average type aggregation operator applied to the sequence $f_{ij}(\cdot)$ [50]. The forgetting factor δ defines the effective memory depth of the weighted averaging aggregation operator. It can be seen that the memory depth K_δ is approximated by the forgetting factor, i.e., $K_\delta = 1/\delta$ [51]. Thus, $F_{ij}(k)$, $F_{oi}(k)$ can be expressed as the weighted average of individual observations on weight W .

Expression (29) can be rewritten to matrix form by replacing $f_{ij}(k)$ and $f_i(k)$ with their matrix/vector counterparts. By encoding the current and next states x_i and x_j as M -dimensional column vectors $\tau(k)$, $\gamma(k)$ where the i th element of $\tau(k)$ is 1 and the rest are 0, and the j th element of $\gamma(k)$ is 1 and the rest are 0. Therefore, $f_{ij}(k) = \tau(k)\gamma(k)^T$, $f_i(k) = \sum_{j=1}^M f_{ij}(k) = \tau(k)\gamma(k)^T 1_M$, where 1_M is an M -dimensional column vector with all elements being 1 [51]. By substituting δ , $\tau(k)$, $\gamma(k)$ into (27), (28), we can obtain the state transfer matrix at moment k as:

$$p(k) = [\text{diag}(F_o(k))]^{-1} F(k) \quad (31)$$

where the matrix $F(k)$ and the vector $F_o(k)$ are formed by $F_{ij}(k)$ and $F_{oi}(k)$, respectively [51].

$$F(k) = F(k-1) + \delta [\tau(k)\gamma(k)^T - F(k-1)] \quad (32)$$

$$F_o(k) = F_o(k-1) + \delta [\tau(k)\gamma(k)^T 1_M - F_o(k-1)] \quad (33)$$

with initial values

$$F(0) = \lambda E; F_o(0) = F(0) 1_M \quad (34)$$

where λ is a small non-negative constant introduced to avoid singularity and E is an $M \times M$ dimensional unit matrix.

Velocity prediction by online updating TPMs

Fig. 9 shows the statistical results of acceleration under Zhengzhou bus driving condition. It can be seen that the acceleration distribution is between -1.8 m/s^2 and 1.3 m/s^2 . The acceleration distribution between $[-1.8, -0.5]$ and $[0.5, 1.3]$ is very sparse and a discrete accuracy of 0.1 m/s^2 is used, while the acceleration distribution between $[-0.5, 0.5]$ is the densest and a discrete accuracy of 0.05 m/s^2 is used. Depending on the range of driving conditions, the grid of velocities is discretized as sets {5,10,15}. The number of grids is obtained by balancing accuracy and computational complexity. In addition, the forgetting factor δ is set as 0.01. The initialization constant λ in (33) is 10^{-12} . The TPMs through online updating every 100 s when the velocity interval is between 18 km/h and 36 km/h are illustrated in Fig. 10.

The predicted velocity result of online updating approach on Zhengzhou bus driving condition when prediction horizon ($H_p = 3 \text{ s}$) is shown in Fig. 11(a). Considering the randomness of future acceleration distribution, probability weighting is applied to the acceleration prediction during velocity prediction. As the updating of TPMs, the predicted velocity distribution tends to be closer to the actual velocity distribution. The absolute error distribution is shown in Fig. 11(b) and the overall RMSE and MAE are 0.1825 m/s and 0.1166 m/s, respectively. Due to the dynamic working condition has a great influence on the degradation of the fuel cell, increasing the prediction horizon will not only reduce the prediction accuracy of the dynamic working condition, but also make the fuel cell degradation cost calculation inaccurate. In order to ensure the details of the dynamic working condition, we need more accurate dynamic working conditions. Therefore, the predicted horizon is chosen as $H_p = 3 \text{ s}$.

Control oriented model

In this study, a model oriented towards linear discrete-time control is considered, with state space functions and system variables defined as follows:

$$\begin{aligned} x(k+1) &= A(k)x(k) + B_u(k)u(k) + B_v(k)v(k) \\ y(k) &= Cx(k) + Du(k) \end{aligned} \quad (35)$$

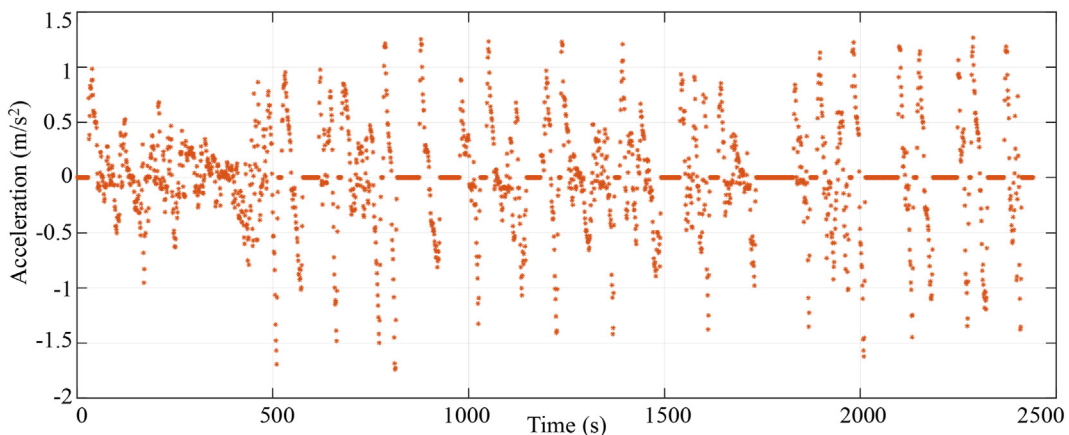


Fig. 9 – Acceleration distribution in Zhengzhou bus driving condition.

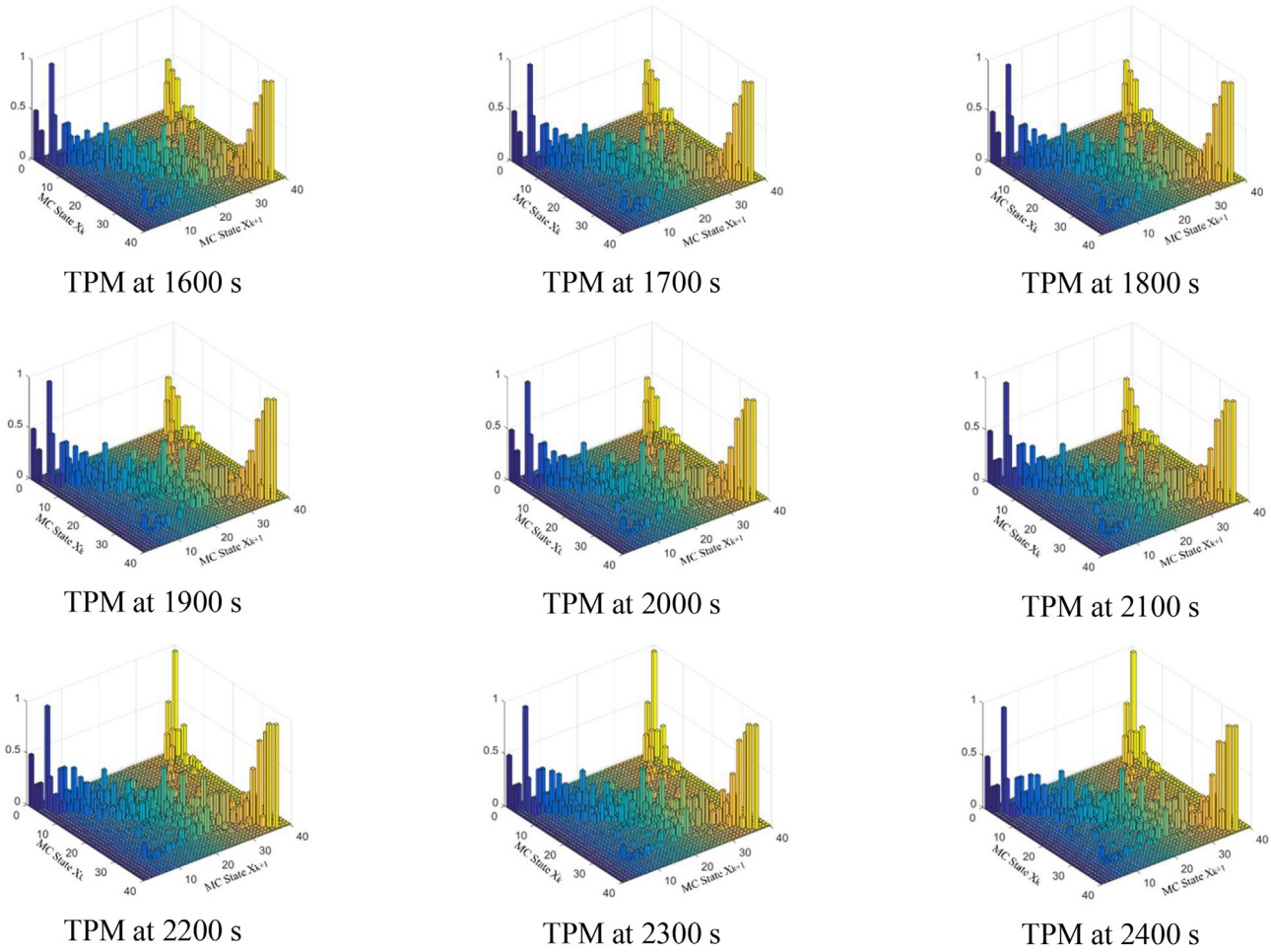


Fig. 10 – Online updating TPMs between 18 km/h and 36 km/h.

In the predictive control strategy, the length of the control horizon is set to be the same as its prediction horizon (H_p) and the sampling period $\Delta T = 1$ s. Where $x(k)$ is the state variable $x(k) = [\text{SoC}(k), P_{fc}(k-1)]^T$, $u(k)$ is the control variable $u(k) = \Delta P_{fc}(k)$, $v(k)$ is the disturbance of the studied system $v(k) = P_{req}(k)$, $y(k)$ is the output variable $y(k) = [\text{SoC}(k), P_{fc}(k)]^T$. Meanwhile, the power balance relationship can be discrete in the following form:

$$P_{req}(k) = P_{fc}(k)\eta_{DC/DC} + P_{bat}(k) \quad (36)$$

In addition, the battery SoC dynamics are modeled by first-order differential equations:

$$\text{SoC}(k+1) = \text{SoC}(k) - \frac{\Delta T}{U_{DC}(k)Q_{bat}} P_{bat}(k) \quad (37)$$

where $U_{DC}(k)$ denote the busbar voltage. According to (34)–(36), the system matrices can be determined as:

$$A(k) = \begin{bmatrix} 1 & \frac{\Delta T \eta_{DC/DC}}{U_{DC}(k)Q_{bat}} \\ 0 & 1 \end{bmatrix} \quad B_u(t) = \begin{bmatrix} \frac{\Delta T \eta_{DC/DC}}{U_{DC}(k)Q_{bat}} & 1 \end{bmatrix}^T \quad (38)$$

$$B_v(k) = \begin{bmatrix} -\frac{\Delta T}{U_{DC}(k)Q_{bat}} & 0 \end{bmatrix}^T \quad C = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad D = \begin{bmatrix} 0 & 1 \end{bmatrix}^T$$

Objective function

In this study, four EMS objectives are considered: (1) hydrogen consumption saving, (2) fuel cells lifetime extension, (3) power cells lifetime extension and (4) battery SoC regulation.

To minimize the total operating cost of the bus in each forecast horizon, the objective function J can be defined as:

$$J_i = C_{H_2,i} + C_{fc,i} + C_{bat,i} + L_{SoC,i}, \quad i = 0, 1, 2, \dots \quad (39)$$

where $C_{H_2,i}$ is the hydrogen consumption cost over the prediction horizon i , i.e.,

$$C_{H_2} = \sum_{m=i}^{i+H_p-1} \psi_{H_2} \dot{m}_f(u(m), v(m)) \Delta T, \quad i = 0, 1, 2, \dots \quad (40)$$

where ψ_{H_2} is the hydrogen price per kilogram, $\dot{m}_f(u(m), v(m))$ is the rate of hydrogen consumption per unit time. In addition, the fuel cell degradation cost $C_{fc,i}$ is computed by:

$$C_{fc,i} = C_{start-stop,i} + C_{idling,i} + C_{change,i} + C_{high-efficiency,i} + C_{high,i}, \quad i = 0, 1, 2, \dots \quad (41)$$

where $C_{start-stop,i}$ denote the degradation cost due to fuel cell start-stop cycles, which can be calculated as:

$$C_{start-stop,i} = \frac{v_1 \cdot n_{cycle,i} \cdot \psi_{fc}}{\Delta V}, \quad i = 0, 1, 2, \dots \quad (42)$$

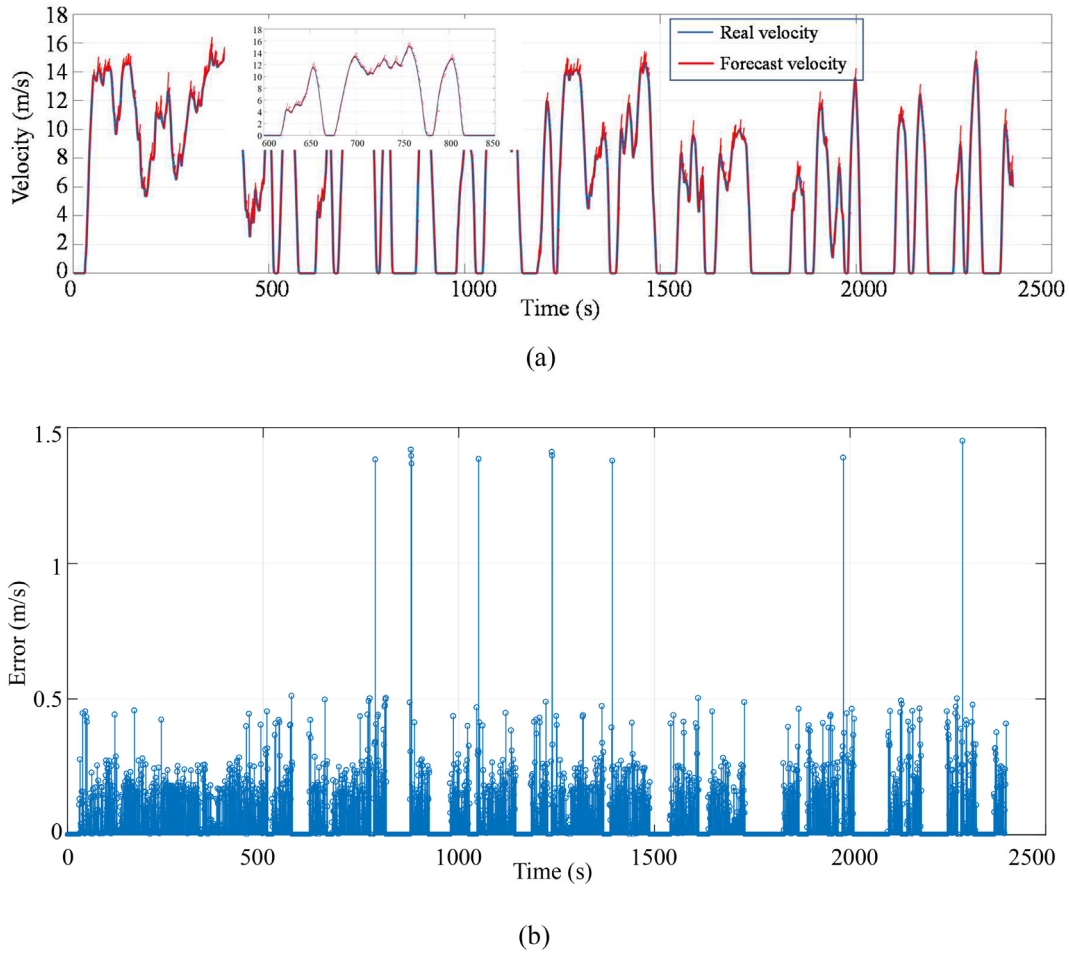


Fig. 11 – The performance of Online updating TPMs on Zhengzhou bus driving condition.

where $n_{\text{cycle},i}$ is the number of fuel cell start-stop cycles over the i th prediction horizon, ΔV is the permitted single fuel cell voltage degradation value from the beginning to end of life-time. Typically, a ten percent voltage drop in a fuel cell at rated current density is considered end of life, and $\psi_{fc} = 60\tau_{fc}$, where τ_{fc} is the PEMFC price per kW. In addition, the cost of degradation caused by low load, as defined by

$$C_{\text{idling},i} = \frac{v_2 \cdot t_{\text{idling},i} \cdot \psi_{fc}}{\Delta V}, \quad i = 0, 1, 2, \dots \quad (43)$$

where $t_{\text{idling},i}$ is the duration of low load in hour over the prediction horizon i . Then, $C_{\text{change},i}$ denote the degradation cost caused by load change, which is computed by

$$C_{\text{change},i} = \frac{v_3 \cdot \sum_{m=i}^{i+H_p-1} \Delta P_{fc}(m) \cdot \psi_{fc}}{n_{fc} \cdot \Delta V}, \quad i = 0, 1, 2, \dots \quad (44)$$

where $n_{fc} = 310$ is the number of PEMFC cells. Meanwhile, the cost of fuel cell degradation due to high efficiency region load and high load is given by the following equation

$$C_{\text{high-efficiency},i} = \frac{v_4 \cdot t_{\text{high-efficiency},i}^1 \cdot \psi_{fc}}{\Delta V} \quad (45)$$

$$C_{\text{high},i} = \frac{v_5 \cdot t_{\text{high},i}^2 \cdot \psi_{fc}}{\Delta V}, \quad i = 0, 1, 2, \dots$$

where $t_{\text{high-efficiency},i}^1$ and $t_{\text{high},i}^2$ are the duration in hour of fuel cell high efficiency region load and high load over the i th prediction horizon.

The degradation cost $C_{\text{bat},i}$ of the battery is calculated as follows

$$C_{\text{bat},i} = (\text{SOH}(i) - \text{SOH}(i + H_p)) \cdot \psi_{\text{bat}}, \quad i = 0, 1, 2, \dots \quad (46)$$

where ψ_{bat} is the cost of battery pack. $\psi_{\text{bat}} = 47.3 \cdot \tau_{\text{bat}}$, where τ_{bat} is the battery price per kWh.

To make the battery charge as continuous as possible, $L_{\text{SoC},i}$ is a deliberately set penalty term, i.e.,

$$L_{\text{SoC},i} = \alpha \sum_{m=i}^{i+H_p} (\text{SoC}(m) - \text{SoC}_{\text{ini}})^2, \quad i = 0, 1, 2, \dots \quad (47)$$

with α is a constant positive weighting factor that should be adjusted by trial and error.

Constraints

To ensure the proper operation of the power sources, complete constraints should be established before making control decisions. First, the constraints on the fuel cell system are expressed as

Table 4 – The cost parameters of hydrogen, PEMFCs and battery unit.

Price ^a	Value
Hydrogen	7.46 USD/kg
PEMFCs	895.52 USD/kW
Battery unit	223.88 USD/kWh

^a Current prices in China.

$$\begin{cases} P_{fc} \leq P_{fc}(k+i) \leq \bar{P}_{fc} & (a) \\ \Delta P_{fc} \leq \Delta P_{fc}(k+i-1) \leq \bar{\Delta P}_{fc} & (b) \\ i \in \{0, 1, \dots, H_p - 1\} & (c) \end{cases} \quad (48)$$

where $P_{fc} = 0$ kW, $\bar{P}_{fc} = 60$ kW, $\Delta P_{fc} = -\bar{\Delta P}_{fc} = 1$ kW/s. Then, the constraints of the battery system are denoted as

$$\begin{cases} \text{SoC} \leq \text{SoC}(k+H_p) \leq \bar{\text{SoC}} & (a) \\ I_{bat} \leq I_{bat}(k+i) \leq \bar{I}_{bat} & (b) \\ \text{SoC}(k+1) = \text{SoC}(k) - \frac{I_{bat}(k)\Delta t}{Q_{bat}} & (c) \\ \text{SoC}(0) = \text{SoC}_{ini} & (d) \\ \text{SOH}(k+1) = \text{SOH}(k) - \frac{|I_{bat}(k)|\Delta t}{2N_{EOL}(C)Q_{bat}} & (e) \\ \text{SOH}(0) = \text{SOH}_{ini} & (f) \\ i \in \{0, 1, \dots, H_p - 1\} & (g) \end{cases} \quad (49)$$

In this study, a battery pack with a nominal energy capacity of 47.3 kW h is used as the energy storage unit for the hybrid powertrain. Where $\text{SoC} = 0.5$, $\bar{\text{SoC}} = 0.62$, $I_{bat} = -200$ A, $\bar{I}_{bat} = 300$ A, $\text{SoC}_{ini} = 0.6$ and $\text{SOH}_{ini} = 1$. Since the vehicle is not equipped with the plug-in characteristics, according to the relationships among OCV, resistance and SOC, the internal resistance of charging and discharging is smaller when the SOC is around 0.6, and the battery efficiency is higher. The $\bar{\text{SoC}}$ is set to 0.62 can ensure that the battery has sufficient charging capacity as well as sufficient discharging capacity.

Optimization solution

With the above objective function and constraints, the power-allocating decision in the i th receding horizon can be expressed mathematically as follows

$$\begin{aligned} \min_{\Delta P_{fc}(k), k=i, \dots, i+H_p-1} J_i &= C_{H_2,i} + C_{fc,i} + C_{bat,i} + L_{\text{SoC},i} \\ \text{s.t. Eq(48) - (49)} \end{aligned} \quad (50)$$

Clearly Eq. (50) is a nonlinear optimization problem with respect to ΔP_{fc} . In the solution, for the optimal control sequence $[\Delta P_{fc}(i), \dots, \Delta P_{fc}(i+H_p-1)]$, only the first element $\Delta P_{fc}(i)$ is applied to the hybrid powertrain model, while the other elements are discarded. As the system state is updated at the next time step, optimization is performed again to find the next control action.

Description of the benchmarking strategies

DP as a globally optimal strategy for a known driving cycle yields minimized vehicle operating cost, as formulated by:

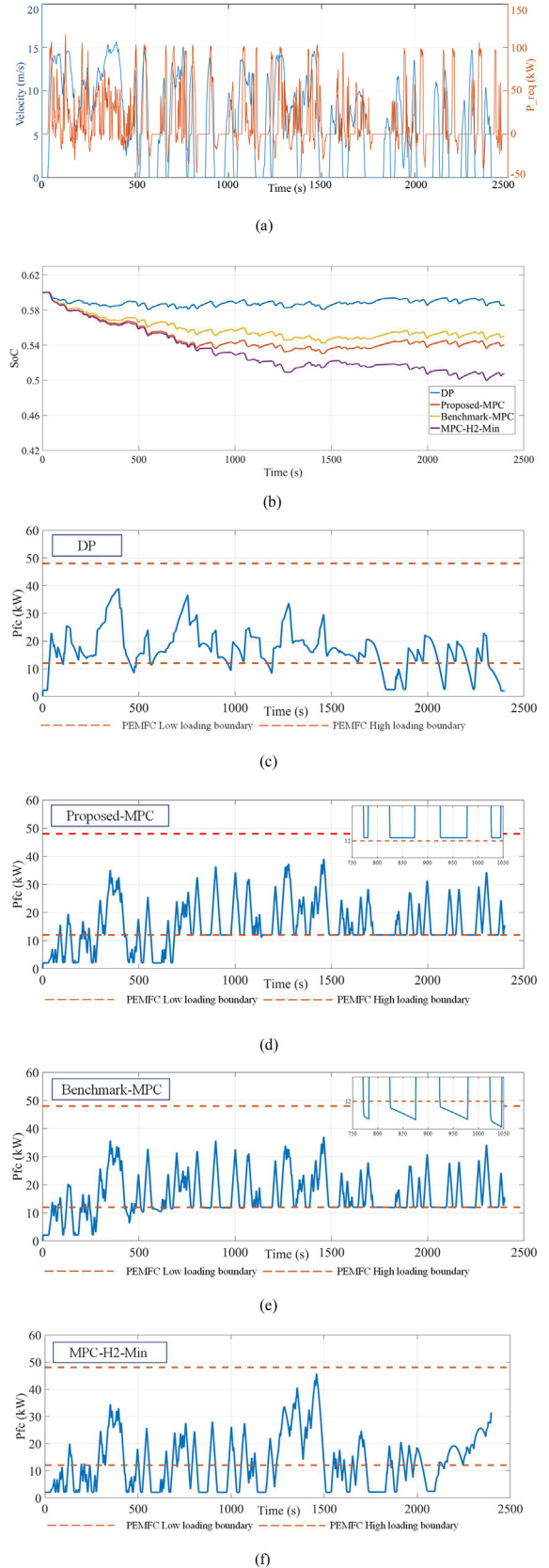


Fig. 12 – The results of three control strategies: (a) Velocity and power demand for Zhengzhou bus driving condition; (b) SoC trajectory of battery; The PEMFCs power profiles of (c) DP, (d) Proposed-MPC, (e) Benchmark-MPC, (f) MPC-H2-Min.

$$\min_{\Delta P_{fc}(i), i=0,1,\dots,N-1} \sum_{i=0}^{N-1} C_{H_2,i} + C_{fc,i} + C_{bat,i} + L_{SoC,i} \quad (51)$$

S.t.

$$\begin{cases} 0kW \leq P_{fc}(i) \leq 60kW & (a) \\ -1kW \leq \Delta P_{fc}(i) \leq 1kW & (b) \\ P_{fc}(i+1) = P_{fc}(i) + \Delta P_{fc}(i) & (c) \\ 0.3 \leq SoC(i) \leq 0.9 & (d) \\ SoC(i+1) = SoC(i) - \frac{I_{bat}(i)\Delta t}{Q_{bat}} & (e) \\ 0 \leq SOH(i) \leq 1 & (f) \\ SOH(i+1) = SOH(i) - \frac{|I_{bat}(i)|\Delta t}{2N_{EOL}(c)Q_{bat}} & (g) \\ SoC(0) = 0.6, SOH(0) = 1, P_{fc}(0) = 0 & (h) \\ 0.4 \leq SoC(N) \leq 0.8 & (k) \end{cases} \quad (52)$$

Constraints (52a)–(52b) and (52c) represent the fuel cell physical operating boundary and dynamic discrete equation. Constraint (52d) and (52e) indicate the SoC boundary and dynamics, respectively. Constraint (52f) and (52g) denote the SOH boundary and dynamic discrete equation, respectively. In

addition, constraint (52h) sets out the initial states of the SoC, SOH and P_{fc} , while (52k) specifies a terminal constraint on the SoC.

Besides, a MPC-based strategy that do not fully consider the whole working regions (Benchmark-MPC) is adopted for comparison, as expressed in Eq. (53), where the detail formulas of $C_{H_2,i}$, $C_{bat,i}$ and $L_{SoC,i}$ are given in Eq. (40), Eq. (46) and Eq. (47). In addition, since the fuel cell operating mainly in the high efficiency region, where $\lambda = 3$ is used as a penalty factor for disregarding the loss of life in the high efficiency region load.

$$\begin{aligned} \min_{\Delta P_{fc}(k), k=i,\dots,i+H_p-1} J_i &= C_{H_2,i} + C_{fc,i}^* + C_{bat,i} + L_{SoC,i} \\ C_{fc,i}^* &= \lambda(C_{start-stop,i} + C_{idling,i} + C_{change,i} + C_{high,i}) \\ \text{s.t.} & \text{Eq(48) – (49)} \end{aligned} \quad (53)$$

In addition, an MPC-based strategy that aims to minimize the cost of H_2 consumption (MPC-H2-Min) is also used as a comparison. The optimization objective is given in Eq. (54), where the detail expressions of $C_{H_2,i}$ and $L_{SoC,i}$ are given in Eq. (40) and Eq. (47).

$$\begin{aligned} \min_{\Delta P_{fc}(k), k=i,\dots,i+H_p-1} J_i &= C_{H_2,i} + L_{SoC,i} \\ \text{s.t.} & \text{Eq(48) – (49)} \end{aligned} \quad (54)$$

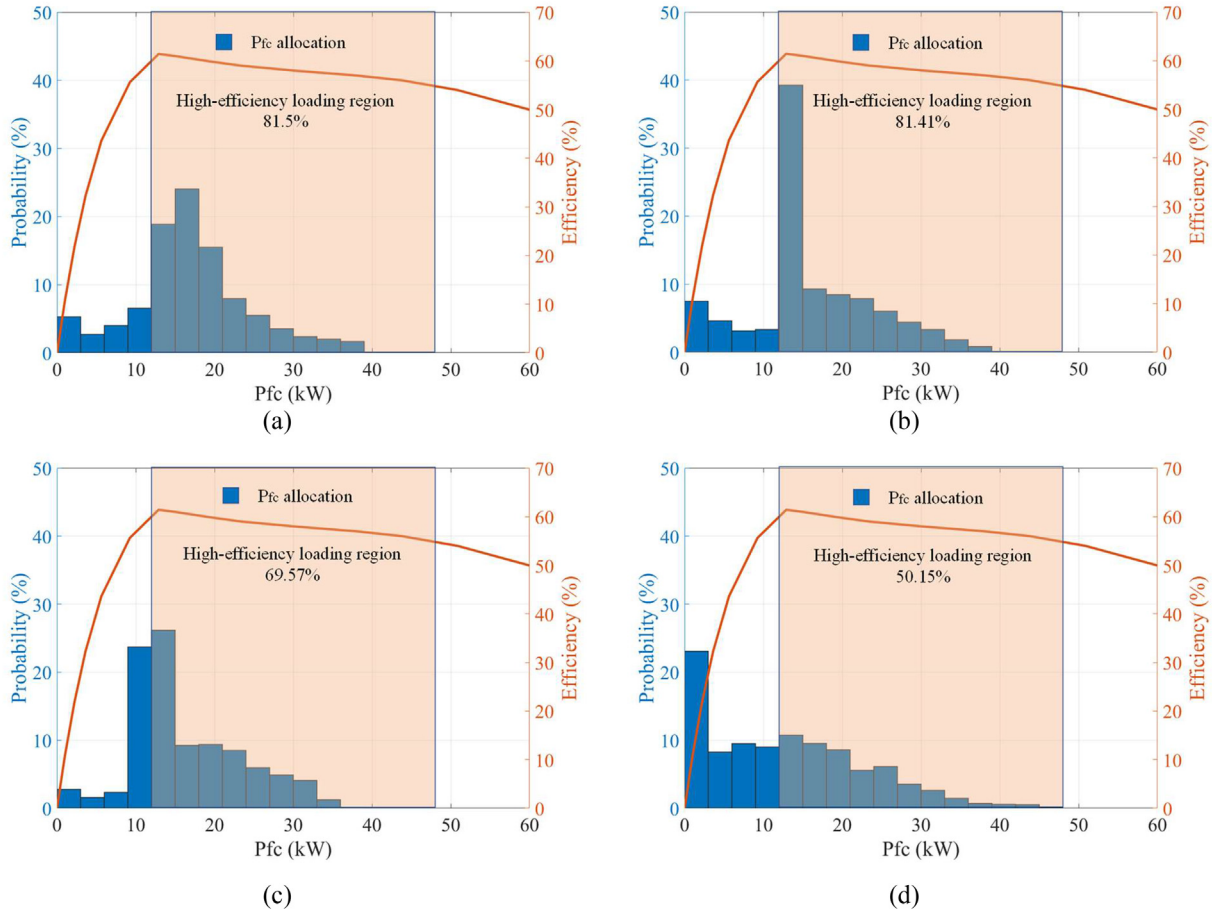


Fig. 13 – Probability allocation of fuel cell output power under different control strategies: (a) DP strategy; (b) Proposed-MPC strategy; (c) Benchmark-MPC strategy; (d) MPC-H2-Min.

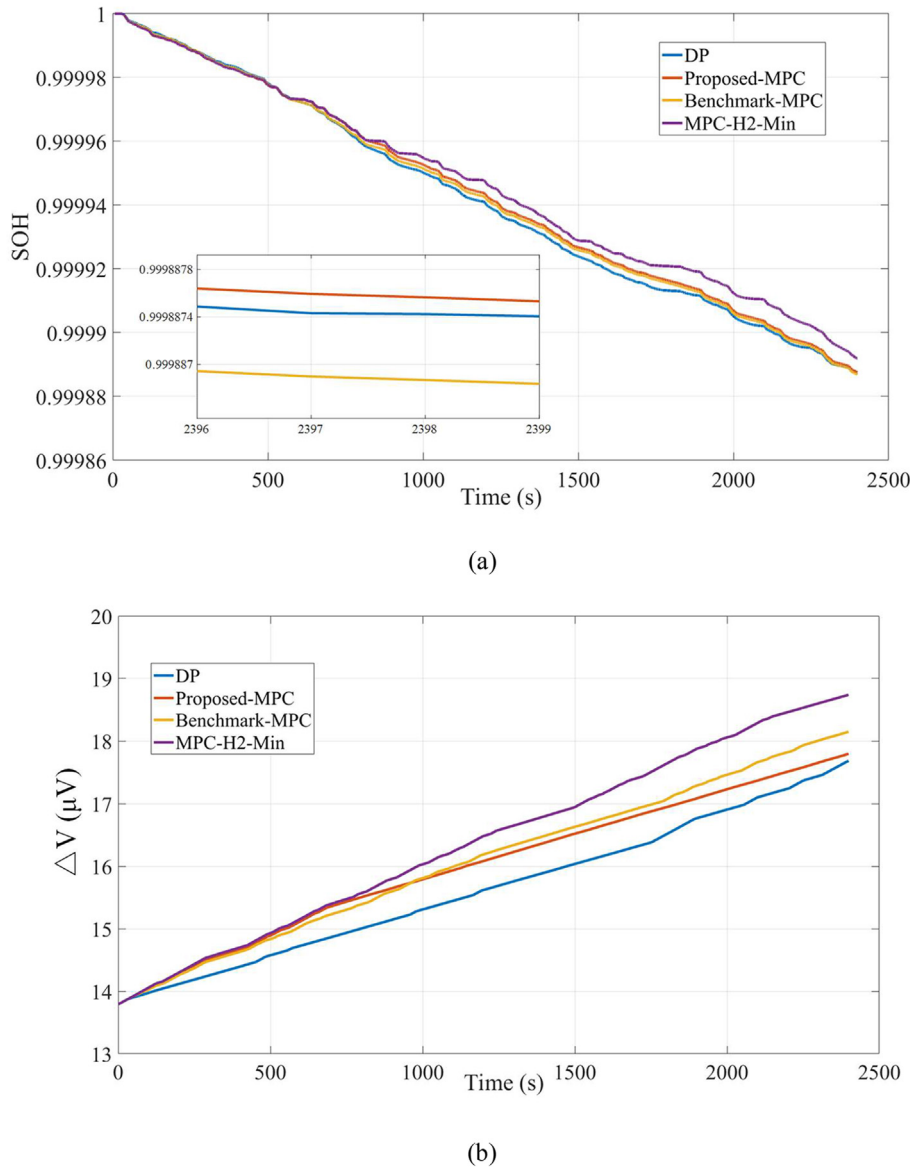


Fig. 14 – Performance under different control strategies: (a) SOH (b) Trajectory of the voltage drop of a single fuel cell.

Results and discussions

In this section, we show the results of controlling the power flow between the fuel cell and the battery in real time under the aforementioned MPC energy management framework. In this paper, the performance of the proposed strategy is compared with the benchmark strategy using the Zhengzhou bus driving condition as a validation to reveal its advantages. The cost parameters of hydrogen, PEMFCs and battery unit are given in Table 4. In this paper, the performance of the EMS is verified using an offline simulation method, which is performed in Matlab/Simulink R2019a with a sampling period of 1 s. The computing device is a desktop computer with an intel core i9-10920X CPU @ 3.50 GHz and a 64G RAM.

To verify the validity of the proposed strategy, we introduce the equivalent hydrogen consumption cost C_{equ} reflecting the SoC variation.

$$C_{equ} = \frac{\Delta SoC \cdot E_{bat} \cdot 3600}{\eta_{fc} \cdot H_{cal}} \psi_{H_2} \quad (55)$$

where ΔSoC represents the variation of SoC, E_{bat} is the nominal energy capacity of the power battery, H_{cal} is the H₂ heating value (120, 000, 000 J/kg).

The performance of four control strategies is compared under Zhengzhou bus driving condition. As shown in Fig. 12 (b), since the complete path information is known in advance, the DP strategy makes the final SoC nearly the same as the initial SoC. In previous studies [30,31] we have known that the MPC-benchmark strategies have achieved good performance in terms of total vehicle operating costs. Two similar SoC profiles are obtained in proposed MPC strategy (Proposed-MPC) and Benchmark-MPC strategy due to the fact that only the cost term $C_{fc,i}$ differs in the two objective functions. In contrast, the MPC-H2-Min strategy prefer to limit the

output power of the fuel cell in order to reduce the cost of hydrogen consumption. This also explains well why the SoC curve based on MPC-H2-Min strategy is below the SoC curve of proposed strategy. In addition, as shown in Fig. 12 (c)–(f), the fuel cell power profile of DP is smoother compared to other strategies. In contrast, the Proposed-MPC and Benchmark-MPC strategies prefer to change the output power of the fuel cell according to the changes of demand power. The main difference between the two MPC-based strategies is that Proposed-MPC strategy tends to make the operating point of the fuel cell work in the high efficiency region as much as possible to reduce the fuel cell degradation. In addition, the distribution of fuel cell output power for the four control strategies under Zhengzhou bus driving condition is given in Fig. 13(a)–(d). In general, PEMFCs operating in the high efficiency region not only improve the overall efficiency of the system, but also reduce the performance degradation caused under other loadings. It can be seen that DP as the optimal baseline strategy makes 81.5% of the fuel cell operating points work in the high efficiency region. The proposed strategy has 81.41% of the operating points working in the high efficiency region, which is better than the Benchmark-MPC strategy (69.57%) and MPC-H2-Min strategy (50.14%).

Fig. 14 (a) depicts the trajectory of SOH under the four control strategies, where Benchmark-MPC strategy has the greatest effect on the battery SOH, followed by the DP, while the MPC-H2-Min strategy has the least effect on SOH. As shown in Fig. 14 (b), DP leads to the least fuel cell unit voltage degradation ($17.6845\mu V$), and compared with Benchmark-MPC strategy, Proposed-MPC strategy can reduce the fuel cell unit voltage degradation from $18.1483\mu V$ to $17.7951\mu V$. Among all strategies, the MPC-H2-Min strategy has the largest fuel cell voltage drop of $18.7391\mu V$.

The composition of the total operating cost for the four strategies is shown in Fig. 15(a)–(d). Where “H₂”, “FC_Start-stop_loss”, “FC_operating_loss” and “Batt_loss” denote hydrogen consumption cost, fuel cell Start-stop degradation cost, fuel cell other operating degradation cost, SOH degradation cost, respectively. Where the results of three of MPC control strategies are corrected. With only one start-stop cycle for the fuel cell in all control strategies, it can be seen that the fuel cell degradation cost accounts for the largest proportion of the total cost, followed by the hydrogen consumption cost. Compared with Benchmark-MPC strategy, Proposed-MPC strategy can reduce “Batt_loss” by 0.62% and “FC_operating_loss” by 8.1%, respectively. Compared with MPC-H2-Min

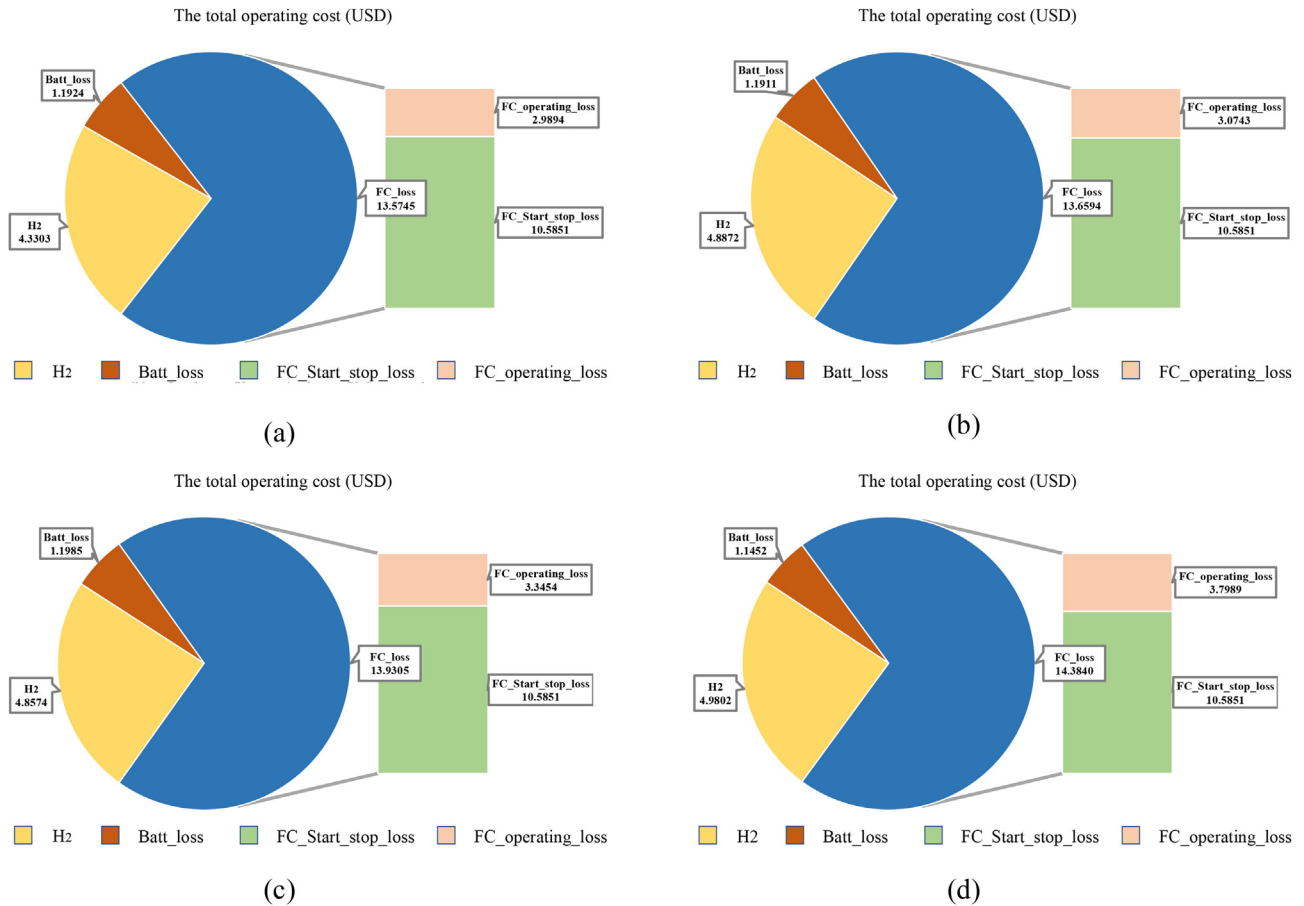


Fig. 15 – Total operating cost of different strategies under Zhengzhou bus driving condition: (a) DP strategy, (b) Proposed-MPC strategy, (c) Benchmark-MPC strategy; (d) MPC-H2-Min.

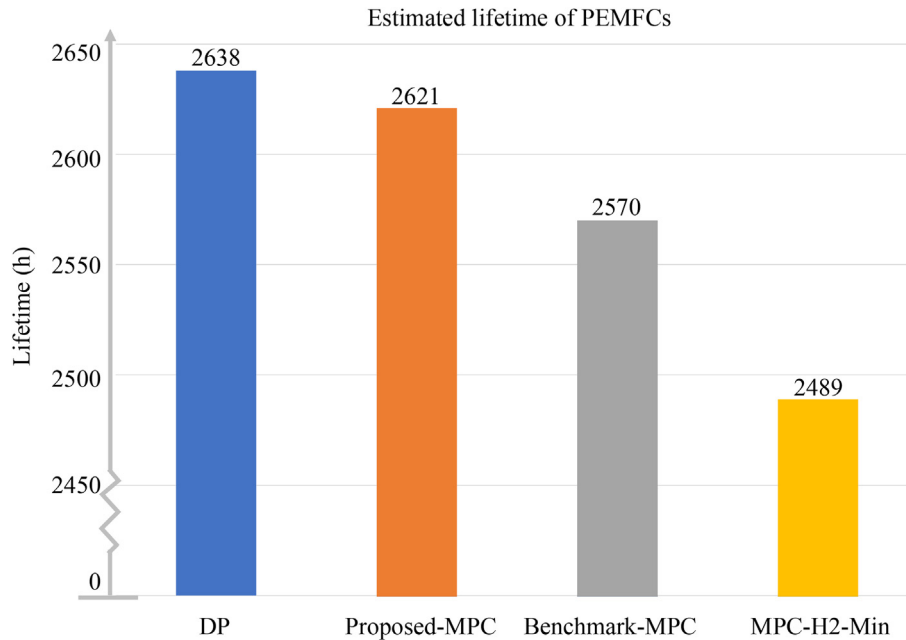


Fig. 16 – Estimated fuel cell lifetime over Zhengzhou bus driving condition.

Table 5 – Comparison of online running time under URBAN cycle condition.

Control strategies	Average running time per step (ms)
Proposed-MPC	90.5
Benchmark-MPC	91.5
MPC-H2-Min	116.5

strategy, Proposed-MPC strategy can reduce “H2” by 1.87% and “FC_operating_loss” by 19.07%, respectively. In summary, Proposed-MPC strategy can reduce total operating cost by 1.24% and 3.76% compared to Benchmark-MPC strategy and MPC-H2-Min strategy, respectively. Compared to DP benchmark, the total operating cost gap is 3.24%.

With the help of Eq. (10b), the estimated fuel cell lifetime T_{fuel} (in hour) for the four control strategies under Zhengzhou bus driving condition is given in Fig. 16. Where T_{fuel} under DP is 2638 h, T_{fuel} under Proposed-MPC strategy is 2621 h, T_{fuel} under Benchmark-MPC strategy is 2570 h and T_{fuel} under MPC-H2-Min strategy is 2489 h. It can be seen that Proposed-MPC strategy can increase the fuel cell lifetime by 51 h (2%) and 132 h (5.04%) compared to Benchmark-MPC strategy and MPC-H2-Min strategy, respectively. In addition, the lifetime optimization gap relative to DP is 17 h (0.64%). The effectiveness of the proposed strategy in prolonging the fuel cell lifetime was confirmed. Table 5 shows the average running time per step for the three real-time energy management strategies. It can be seen that the MPC-H2-Min strategy consumes the largest amount of calculation time. Although the running time per step of Benchmark-MPC strategy is a little larger than that of Proposed-MPC, it is smaller than the average sampling period (1 s). Therefore, this implies its potential for real-time applications.

Conclusions

In this paper, a fuel cell hybrid electric bus energy management strategy with fuel cell/battery health awareness control is proposed to minimize the operating cost of FCHEB based on a MPC framework. First, the effects of different loading conditions on the fuel cell lifetime are analyzed, and the optimal control sequence is solved by DP with the help of an improved online Markov predictor. Then, the effectiveness of the proposed strategy is verified by comparing the analysis with multiple benchmark strategies. The main conclusions are as follows:

- (1) Compared with the other two MPC strategies, the Proposed-MPC strategy optimizes the output power distribution of the fuel cell and make 81.41% of the operating points working in the high efficiency region, which is better than 69.57% of the Benchmark-MPC strategy and 50.15% of the MPC-H2-Min strategy. In addition, the estimated lifetime of the fuel cell with the proposed strategy is 2621 h, which can increase the fuel cell lifetime by 51 h (2%) and 132 h (5.04%) compared to Benchmark-MPC strategy and MPC-H2-Min strategy, respectively, thus verifying the effectiveness of the proposed strategy in prolonging the fuel cell lifetime.
- (2) Under Zhengzhou bus driving condition, compared with Benchmark-MPC strategy, the proposed strategy can reduce the fuel cell degradation cost by 1.95%, although the advantage in hydrogen consumption is not obvious. Compared with MPC-H2-Min strategy, Proposed-MPC strategy can reduce the hydrogen consumption by 1.87% and fuel cell degradation by 5.04%, respectively. In addition, the total operating cost gap with DP is 3.24%.

Credit author statement

Hongwen He: Investigation, Methodology, Software. Chun-chun Jia: Software, Investigation, Writing-original draft preparation. Jianwei Li: Investigation, Conceptualization, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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